UPSC Prelims

Science & Tech

Class No. 13

Quantum Computing

Quantum Computing

- Quantum effects : we read in nanotech terms
- Quantum scale : scale at atomic level
- What is the difference?
- At the quantum level, laws of classical physics and mechanics do not apply.
- Quantum computing is the area of study focused on developing computer technology based on the principles of quantum theory, which explains the nature and behaviour of energy and matter on the quantum (atomic and subatomic) level.

Classical Computing

- classical computers operate using binary physical state, meaning its operations are based on one of two positions (1 or 0).
- 1 : electricity is being passed through the wire
- 0: no electricity is being passed through the wire
- Classical computers process information in a binary format, called bits, which can represent either a 0 or 1.

Classical Computing

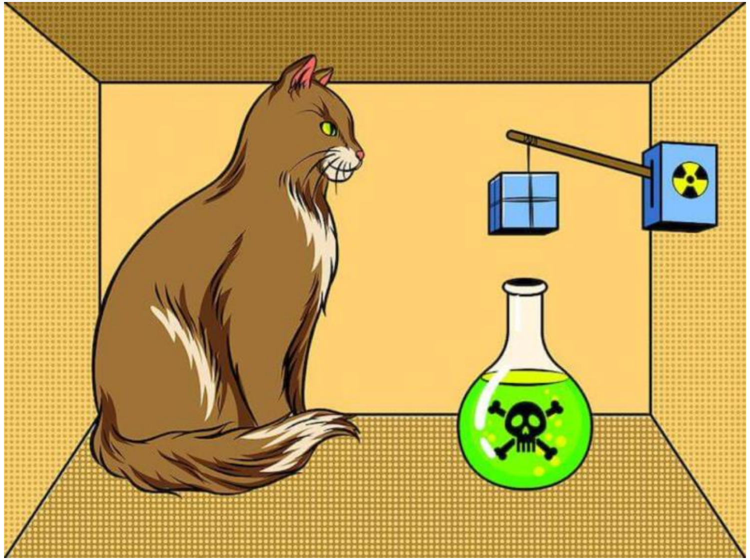
- Quantum computers, in contrast, use logical units called quantum bits, or qubits for short, that can be put into a quantum state where they can simultaneously represent both 0 and 1 and their correlations.

2 key principles of Quantum Computing

- Superposition
- Entanglement
- Let's understand these 2 in detail

Superposition : Schrodinger's cat

- Is the cat dead or alive?



Superposition : Schrodinger's cat

- The cat is both dead and alive at the same time
- Only when you open the box will you know if it is dead or alive
- This thought experiment is used to understand the quantum state of electrons.
- Superposition means that each quantum bit (basic unit of information in a quantum computer) can represent both a 1 and a 0 at the same time.

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

2nd Property : Entanglement

- In quantum entanglement, subatomic particles become “entangled” (linked) in such a way that any change in one disturbs the other even if both are at opposite ends of the universe.

Clarity for UPSC by Dr. Shivin

Ed Sarrthi
Your Own Mentor

Let me take you to a mcdonald's drive through

- You order a mc chicken and a mc veggie
- However, they give you both the burgers in an exact same box.
- Now, when you open the first box, and you see a mc chicken, you know the other box has a mc veggie.
- The state of both these burgers is entangled : change in one means the change in other.

Entanglement in Quantum Computing

- Here we are talking about electrons at subatomic level
- Electrons have a spin : either positive or negative
- They aren’t actually spinning : this is to understand their orientation
- Superposition of spin : think?
- Imagine a coin being flipped on a table : demonstration
- When there is interaction with the outside world : there is collapse of the wave function and electron shows its spin

Entanglement

- 2 electrons close to each other : they will get opposite spin
- Their spin states will become entangled / linked
- Let’s watch a video and understand

Entanglement leads to Superlinkages

- Despite a huge / astronomical distance, disruption in one electron will lead to a change in the other electron
- This has huge applications in Communication and Cryptography
- In quantum entanglement, subatomic particles become "entangled" (linked) in such a way that any change in one disturbs the other even if both are at opposite ends of the universe

Quantum Computing

- Recap :
- 2 principles : Superposition and Entanglement
- Much Faster than Other Computers
- Quantum principles will be used for engineering solutions to extremely complex problems in communications, sensing, chemistry, cryptography, imaging and mechanics.

Basics

- Qubits: basic unit of quantum computing; can contain both values “1” and “0” simultaneously.
- Superposition: ability to be in multiple states simultaneously → provides exponential speed.
- Entanglement: ability of two members of a pair (Qubits) to exist in a single quantum state.

Basics

- Quantum supremacy: state when a quantum device can solve a problem that no classical computer can solve in any feasible amount of time (irrespective of the usefulness of the problem).

Advantages of Quantum Computing

- **Speed**: perform quick and easy calculations that are time consuming on conventional computers.
- **Storage**: drastically bring down the space needed to store data.
- **Simulation**: doing data simulation involving multiple parameters like weather forecasting, chemical simulation.
- **Privacy**: impossible to break the security of quantum entanglement-based cryptography

Applications

- <u>**Secure Communication**</u>: quantum cryptography → significant to satellites, military and cyber security; promises unimaginably fast computing and safe, unhackable satellite communication to its users. E.g. China demonstrated secure quantum communication between terrestrial stations and satellites; E.g. March 2021 ISRO demonstrated free space Quantum communication.

Applications

- **Research**: solving some of the fundamental questions in physics related to gravity, black hole etc; could give a big boost to the Genome India project. E.g. CERN (God particle)
- **Agriculture**: precision agriculture – predicting rainfall, type of soil, crop and fertilizers; make precise long-term large-scale production plans and helps to increase yield and efficiency; detecting weed through an invasive weed optimization algorithm.

Applications

- **Pharmaceutical**: Enhanced data processing abilities - develop vaccines faster, predict epidemics, simulate drug/vaccine responses/side effects, genome analysis and gene-editing; could reduce the time frame of the discovery of new molecules; E.g. Covid 19 variants

Applications Continued

- **Augmenting Industrial revolution 4.0**: Quantum computing is an integral part of Industrial revolution 4.0.; help leveraging other IR 4.0 technologies like the Internet-of-Things, machine learning, robotics, and AI.
- **Artificial intelligence and big data**: could empower machine learning by enabling AI programs to search through the gigantic datasets, consumer behavior, financial markets, etc.

Applications Continued

- **Weather: Forecasting and Climate Change**: Enhancing weather system modeling allowing scientists to predict the changing weather patterns with excellent accuracy; meteorologists can generate and analyze more detailed climate models - will provide greater insight into climate change and ways to mitigate it.
- **Disaster Management**: collection of data regarding climate change can be streamlined; overlaying this weather modelling data; Tsunamis, drought, earthquakes and floods may become more predictable

Quantum Supremacy

- It's the point at which a quantum computer can complete a mathematical calculation that is beyond the reach of even the most powerful supercomputer.
- Recently, Sycamore (Google's quantum computer) took 200 seconds to perform a calculation that the world's fastest supercomputer, Summit, would have taken 10,000 years to do.

Challenges

- **Quantum physics** – programmers will also need to know and learn particle physics;
- **Quantum states** – Quantum decoherence : the Qubits loose the quantum state easily
- **Designing quantum algorithm** – Need specialized algorithm, totally different from conventional algorithm.
- **Cyber Security**: QC can decrypt all the data on cyber space
- **Limited working**: The tasks like word processing and email are not suitable for quantum computers.

Quantum Key Distribution

- In traditional cryptography, security is based on that an adversary is unable to solvea certain mathematical problem.
- In QKD, security is achieved through laws of quantum physics.
- 2 most important laws are: Superposition and Entanglement.

Quantum Key Distribution

- News Context: A satellite-based communication between 2 ground stations was activated by entangled-based quantum key distribution (OKD).
- This was achieved by Micius (Quantum Experiments at Space Scale), World's 1St quantum-enabled satellite.
- Micius was launched by China in 2016

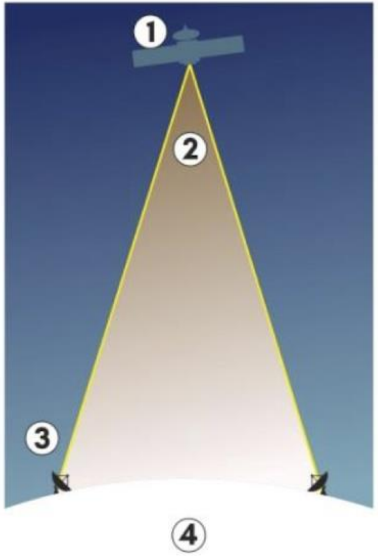
Quantum Key Distribution

- News Context: Bengaluru based startup Qnu labs recently innovated advanced secured communication through QKD.
- Qnu labs under aegis of Innovation for Defence Excellence (iDEX)
- iDEX is the operational framework of Defence Innovation Organization, a special purpose vehicle under Ministry of Defence.

How does Quantum Key Distribution works?

Quantum key distribution allows user to agree on a way of transmitting their data without the worry that someone is listening in

1. Sender instructs satellite to generate 2 entangled photons of particular quantum state
2. Photons are beamed to both ground stations
3. Sender and receiver compare the quantum states of the photons to check if they have been intercepted. If not they use the photons to create a code to encrypt the date
4. Encrypted data can then be sent securely via conventional means



Quantum Key Distribution

- Two main categories of QKD are:
- Prepare-and-measure protocols : focus on measuring unknown quantum states. This type of protocol can be used to detect eavesdropping, as well as how much data was potentially intercepted.
- Entanglement-based protocols focus on quantum states in which two objects are linked together, forming a combined quantum state. In this method, if an eavesdropper accesses a previously trusted node and changes something, the other involved parties will know.

Quantum Key Distribution

- Quantum cryptography is considered future proof : it is believed that no future changes in technology can crack quantum cryptography.

QKD Challenges

- Transmission Loss
- Lower Communication rates
- Infrastructure shortfalls

Quantum Computing and India

- National Mission on Quantum Technologies and Applications (NM-QTA): Budget 2020 allocated Rs 8000 Crore to the mission for a period of five years.
- Quantum Information and Computing (QuIC) lab at the Raman Research Institute, Bangalore.
- Quantum-Enabled Science & Technology (QuEST) a research program by the Department of Science & Technology.

Quantum Computing and India

- Quantum Frontier mission: It is an initiative of the Prime Minister's Science, Technology, and Innovation Advisory Council (PM-STIAC) which aims to initiate work in the understanding and control of quantum mechanical systems.
- DRDO Young Scientist Laboratory for Quantum Technologies (DYSL-QT), Mumbai.

Supercomputers

- Supercomputers are large systems that are specifically designed to solve complex scientific and industrial challenges.
- Unlike traditional computers, supercomputers use more than one central processing unit (CPU).
- Supercomputing is measured in floating-point operations per second (FLOPS).
- Petaflops are a measure of a computer's processing speed equal to a thousand trillion flops.

Supercomputers

- Floating-point is a method of encoding real numbers within the limits of finite precision available on computers.
- Supercomputers can be one million times more processing power than the fastest laptop
- They work on parallel computing or parallelism where many processors work together.
- World Fastest Supercomputer is Fugaku of Japan with speed of 415 Peta-Flops

Supercomputers

- It is a common benchmark measurement for rating speed of microprocessors.
- For comparison, a desktop computer has
- performance in the range of hundreds of giga FLOPS to tens of tera FLOPS.
- 1 Mega FLOPS: Equal to 1 million FLOPS.
- 1 Giga FLOPS: Equal to 1 billion FLOPS.
- 1 Tera FLOPS: Equal to 1 trillion FLOPS.
- 1 Peta FLOPS: It can be measured as 1.000 teraflops.

National Supercomputing Mission

- NSM, launched in 2015, is being steered jointly by Ministry of Electronics & Information Technology and Department of Science and Technology and implemented by C-DAC and Indian Institute of Science, Bangalore.
- Mission envisages empowering our national academic and R&D institutions spread over the country by installing a vast supercomputing grid comprising of more than 70 high-performance computing (HPC) facilities.

National Supercomputing Mission

- Till now C-DAC has deployed 11 such facilities under NSM Phase-1 and Phase-2 with a CCP of more than 20 Petaflops. Eg: PARAM Shivay [IIT (BHU)], PARAM Shakti (IIT Kharagpur), PARAM Brahma (IISER, Pune) etc.
- These supercomputers will also be networked on the National Supercomputing Grid over National Knowledge Network (NKN).
- 4 major pillars of NSM are Infrastructure, Applications, R&D, Human resource development.

National Supercomputing Mission

- NSM is being used for Genomics and Drug Discovery, Urban Modelling, Flood Early Warning and Prediction System for River Basins, seismic Imaging to aid Oil and Gas Exploration etc.
- Recently : PARAM Ganga

India's Supercomputers

- Pratyush : Located in Indian Institute of Meteorology.
- Mihir: Located in National Centre for Medium Range Weather Forecasting.
- PARAM Siddhi : It will be India ' s largest High-Performance Computing and Artificial Intelligence (HPC-AI) supercomputer. The supercomputer will have speed o f 210 AI Petaflops.
- Param Shakti and Param Brahma: Installed at IIT-Kharagpur and IISER, Pune.

Blockchain Technology and Cryptocurrency

- Blockchain : a chain of blocks that have certain information
- Decentralised Ledger.

Cryptocurrency and Blockchain

- Think : why can money not be digitized ?

Cryptocurrency and Blockchain

- The internet is centralised
  - Eg: Google.
  - Trust deficit
- Internet has abundance : no scarcity
- Hacking might be possible.
- These 3 problems are why money cannot be digitized

Sol = Blockchain

Normal Transaction

Akshay

₹100

Ashima

₹30

Pranjal

₹50

Record → Ledger

Gunjan

- Eventually, central authority
- won't do it for free

How to Decentralise?

- We will maintain the ledger ourselves.
- How?

How to Decentralise?

- Ashima pays Akshay Rs. 100
- Akshay pays Pranjal Rs. 30
- Pranjal Pays Ashima Rs. 10

At the end of the month you get together and tally.

How to Decentralise?

- However , if you start with a certain amount
- You won’t need to tally till you don’t overspend.

How to Decentralise?

- Akshay has Rs. 100
- Ashima has Rs. 100
- Pranjal has Rs. 100

- Ashima pays akshay Rs. 100
- Akshay pays pranjal Rs. 30
- Pranjal pays Ashima Rs. 10

- Ashima pays Pranjal Rs. 50

- To insure this you need record of all transactions

How to Decentralise?

- Each set of transaction - Recorded on a block.
  - You need record of previous transactions to validate future ones.

Chain of blocks

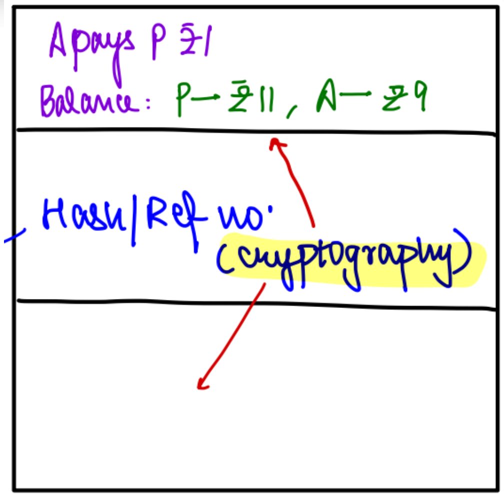
How to Decentralise?

Akshay (A) has Rs. 10

Pranjal (P) has Rs. 10

This is the fingerprint of that block

Change data in any compartment the hash changes.



Cryptography : one way function

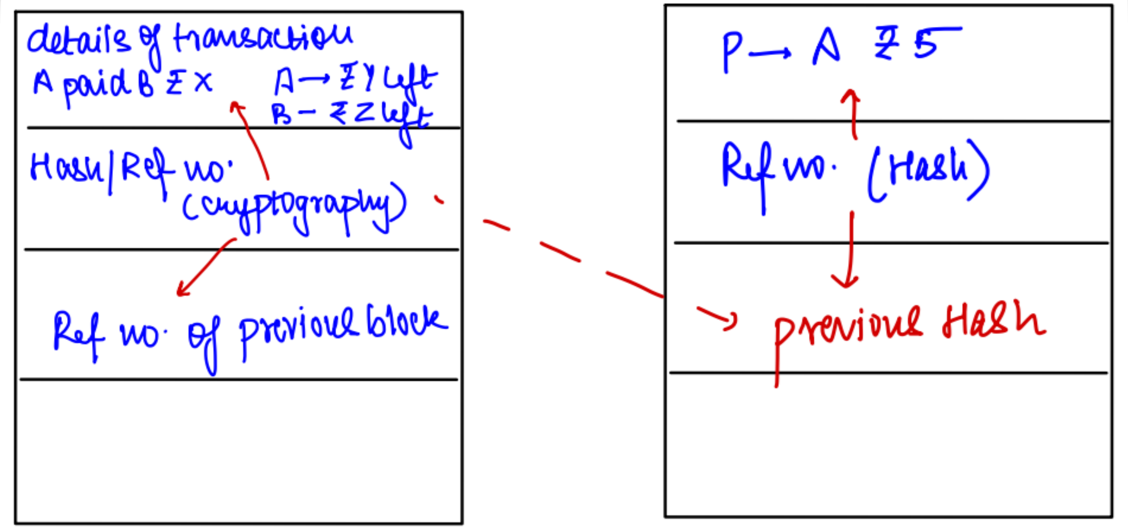
A pays P ₹1

Balance: P → ₹11, A → ₹9 ⟶ 1471329897425

New Transaction :

- Your friends will check the previous ledger
- Validate
- Add to the chain

How to link the new block to the previous one ?

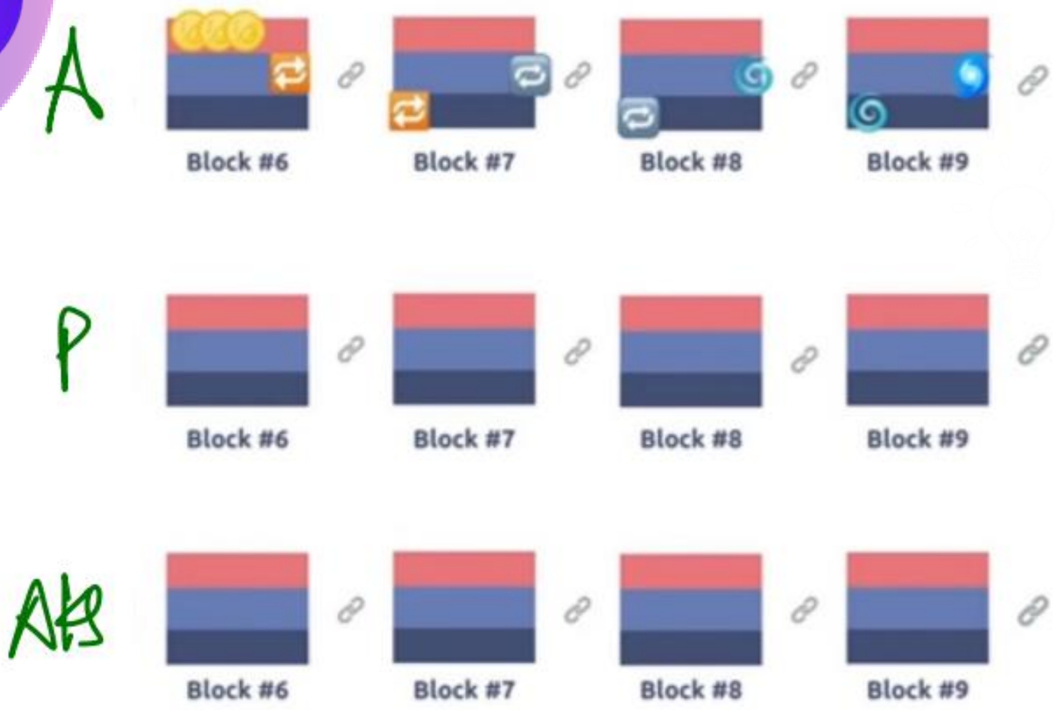


Block - Chain

- It exists on your computer. Not on the servers of a centralized tech giant like google. : Decentralised
- What is a block chain? : Record of transactions : Ledger
- Therefore, Block chain is a decentralized Ledger system.

Block

- Hash (#)/ref no. (unique fingerprint);
  - Generated on basis of 1 and 3 compartment
- New transaction
  - If any illegitimate content : Ref # will change completely
  - Validated - added to chain of blocks.
- Block chain
  - Not centralised
  - Everyone has copy, and any one can validate (What if you change a block) ?



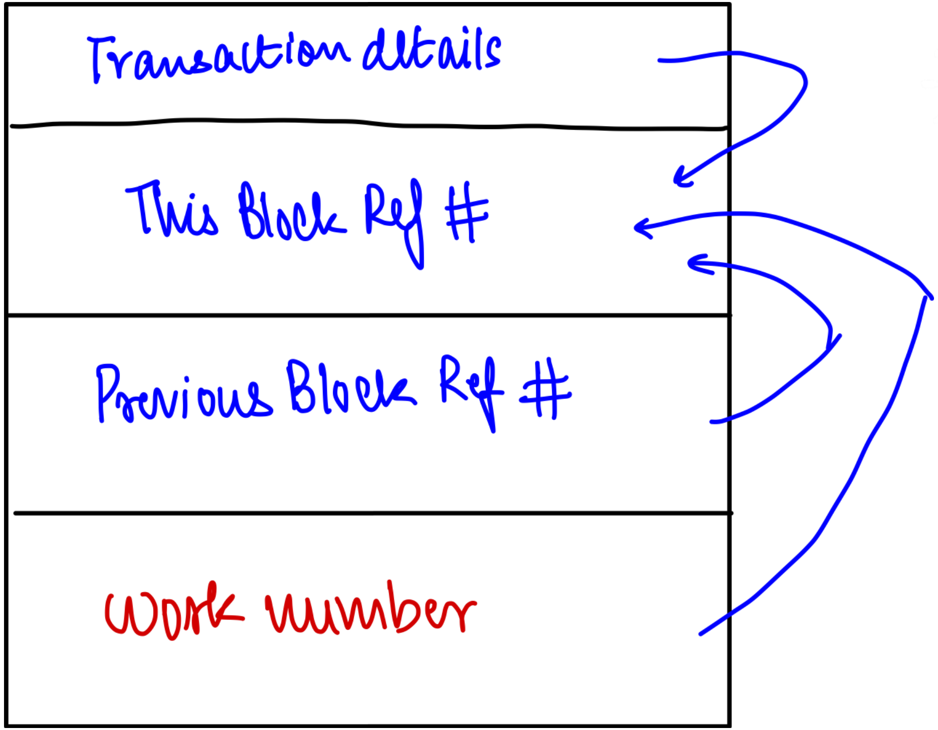
- Decentralisation
  - Change Resistant
  - Hack Proof

Blockchain - Decentralised

- Since all are linked you can't randomly change any record.
  - Change resistant Hack proof.
- Can be validated/invalidated by anyone
  - Digital scarcity + Decentralization
- Missing = Accountability

What if 10000 people blockchains simultaneously

- You will end up with 10000 different chains
- Work needs to be done to change any block chain.
- Changing anything becomes tough


Transaction details
This Block Ref #
Previous Block Ref #
Work number

What if 10000 people blockchains simultaneously

- Lets say rule that all Ref # must start with 0000
- To get 0000 - manually center no. from 1 to 1 million, at some point you'll get 0000
- Lot of computational power required Because work number of all block used to change.
- If before this someone sees that ref# of last block is off : it has no zeroes in the beginning, there would be a problem.

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

What if 10000 people blockchains simultaneously

- Moreover, new blocks are constantly being added. Therefore it becomes very difficult to change a block chain.
- The longest chain would obviously be more secure.

What if 10000 people blockchains simultaneously

- New transactions need to be validated. Why would someone validate transactions / remain a good actor?
- Validation here is calculation of work number for a particular block. This is also known as mining.
- Reasons for remaining a good actor :

  1. You need a lot of computational power for validation.

  2. Every time you validate new currency : you will get tokens. These tokens have value only till the time your chain has validity.

Validation: Trust needs to be established

- Proof of work
- Proof of stake
- Proof of authority

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

Proof of work : Bitcoin

- Proof of work means calculation of work number for validation of new transactions. It is a highly complex problem requiring lots of computational power and electricity. You will get certain tokens for demonstrating proof of work.
- Problem : every validation is a competition between miners, therefore lots of computational power gets wasted. This leads to environmental problems / water wastage in cooling systems.

Proof of work : Bitcoin

- 51 % attack : if one actor has more computational power than all the others in the system, (rest have 49%) then this actor can choose to include or exclude any transaction it wants.
- This person can solve the computational puzzle faster than all others and this person will constantly have the longest chain.

Proof of stake : ethereum

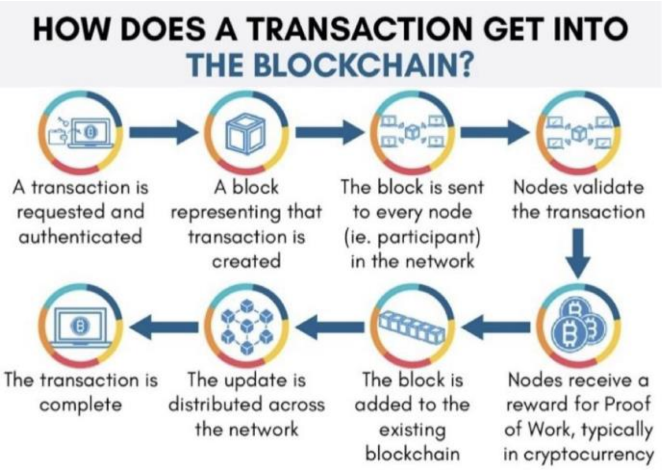
- Every person who wishes to validate a new transaction needs to put certain tokens they have on stake. Usually there is no competition as the block creator is the one who has maximum tokens.
- Since almost always the same person gets to validate, there is no reward for making a block, so the block creator earns only a transaction fee.
- PoS replaces miners with validators and rewards only the top stakeholders, resulting in less energy consumption.

Proof of authority

- Consensus protocol is based on value of identity. There are limited number of moderators and it is close to a centralized system.
- Reputation based system.
- No mining / validation.

Recent Development

- Ethereum switched to merge software mechanism that uses proof of stake (PoS) mechanism. This will help slash energy requirements.
- Ethereum is a decentralized blockchain platform used to build decentralized Apps and Smart Contracts as well as a cryptocurrency.
- From proof of work to proof of stake consensus mechanism.



	Proof-of-Work (PoW)	Proof-of-Stake (PoS)
About	Mechanism bitcoin uses **to regulate creation of blocks** through the process of mining.	An alternative consensus mechanism that **delegates control of the network to the owners of a given token.**
Working Mechanism	**PROOF OF WORK** To add each block to the chain, miners must compete to solve a difficult puzzle using their computers processing power. In order to add a malicious block, you'd have to have a computer more powerful than 51% of the network. The first miner to solve the puzzle is given a reward for their work.	**PROOF OF STAKE** There is no competition as the block creator is chosen by an algorithm based on the user's stake. In order to add a malicious block, you'd have to own 51% of all the cryptocurrency on the network. There is no reward for making a block, so the block creator takes a transaction fee.
Energy Consumption	**PoW consumes more energy** since it allows all miners on a network to try and validate a transaction.	**PoS replaces miners with validators** and rewards only the top stakeholders, **resulting in less energy consumption.**

Blockchain

- Blockchain is a distributed or decentralised ledger technology which was first introduced in the design and development of cryptocurrency, Bitcoin in 2009.
- Blockchain technology is an amalgamation of various technologies such as distributed systems, cryptography, etc
- Blockchain is a series of blocks, where each block contains details of transactions executed over the network, hash(address) of the previous block, timestamp etc.

Blockchain

- Data and transactions stored in blocks are secured against tampering using cryptographic hash algorithms and are validated and verified through consensus (consensus protocols) across nodes of the Blockchain network.
- Blockchain platforms are being developed to offer Blockchain based digital transaction platforms. Popular blockchain platforms include Hyperledger, Cosmos, Polkadot, Redbelly, Ethereum etc.

BlockChain : National Strategy

- National Blockchain Framework : strategy for the next 5 years
- **Need:** Reports predicts that by 2030, Blockchain would be used as a foundational technology for 30% of the global customer base.
- **Plan:** MeitY has initiated a project on design and development of a **NBF** for creation of a shared Blockchain infrastructure and offering Blockchain as-a-Service (BaaS). Initially, NBF would be used for e-governance domain transitioning to incorporate various use cases over time.


Objectives and Vision of National Strategy on Blockchain
To create trusted digital platforms through shared Blockchain infrastructure
Promoting research and development, capacity, skill and innovation, technology and application development
Making India a global leader in Blockchain Technology
Facilitating state of the art, transparent, secure and trusted digital service delivery to citizens and businesses

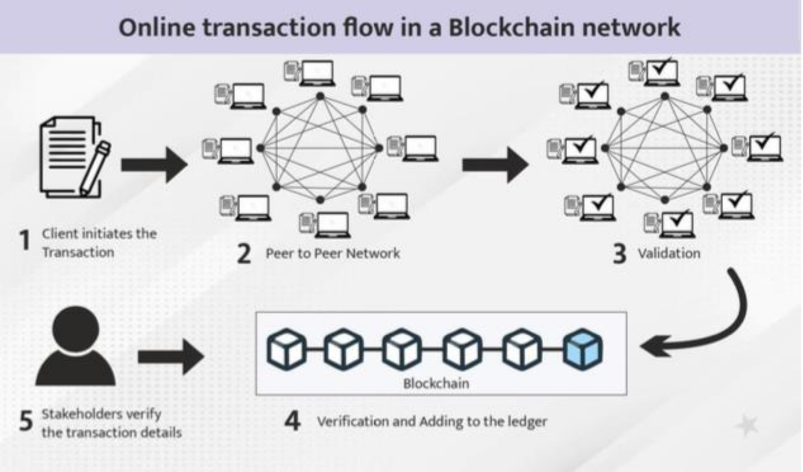
Blockchain National Strategy

- Components :
- Geographically distributed nodes
- R and D
- Indigenous blockchain platform
- Awareness creation
- Integration with online services such as digi locker
- Awareness Creation

Blockchain : Related news

- Presidio Principles : Foundational Values for a decentralized future : released by world economic forum
- 4 pillars : transparency and accessibility, privacy and security, accountability and governance, and Agency and interoperability
- First country to give legal tender status to bitcoin : El Salvador

Blockchain : Potential Applications



Potential Blockchain Applications	
• **Transfer of Land Records** (Property Record Management); • **Digital Certificates Management** (Education, Death, Birth, agreements, etc); • **Pharmaceutical supply chain;** • **e-Notary Service** (Blockchain enabled e-Sign Solution); • **Farm Insurance;** • Identity management; • **Power distribution;**	• Duty payments; • Agriculture and other supply chains; • **eVoting;** • Electronic Health Record Management; • **Digital Evidence Management System;** • Public Service Delivery; • IoT Device Management and Security; • **Vehicle lifecycle management;** • **Chit fund operations** administration; Microfinance for Self-Help Groups (SHG)

Blockchain : Potential Applications

- Currency : crypto
- Banking and Finance : Transaction ledgers
- Inventory : Supply Chain Management
- Health : patient records : digital health mission
- Property and ownership records
- Smart contracts : update regularly
- IPR and asset protection
- Good governance : India Chain
- Corruption : audits

Blockchain : Cryptocurrency

- Cryptocurrency → digital currency secured by cryptography → medium of exchange where the ownership data is stored in a computerized database; Based on→ Blockchain technology
- Cryptography- The science of protecting information by transforming it into a secure format through encryption.
- Decentralized; Security; Anonymity; Speed; Ease of transfer; Cross border trade;
- Has certain advantages and disadvantages

Blockchain : Cryptocurrency

- **Wallet**- the account addresses of the user which is accessible only by a private and a public key; the private key is unique and known only to the owner of the wallet.
- Instantaneous transaction →help save time of both the parties; everything is being done on the internet.
- International monetary market : work in countries with unstable economy; poor institutional strength;

Blockchain : Cryptocurrency

- Disadvantages
- Vulnerability: It's hidden nature of transaction →illegal activities → terror-funding, money laundering and tax-evasion.
- The RBI & Ministry of Finance issued statements on cryptocurrencies ;
- Frauds and scams – Squid cryptocurrency
- Payments in one go: Payments cannot be reversed back; no rollback and no central legal authority;

Blockchain : Cryptocurrency

- Acceptance: Not legal in many countries → holds different values at different places; almost 90% of the currencies are scams.
- Risky investment: Not issued by the government agency of most countries→ makes it less reliable;
- Reports of hacking of several exchanges; Highly volatile;
- Price: depends on demand and supply; high volatile; cannot be relied as medium of exchange of value

Blockchain : Cryptocurrency

- Disadvantages in obtaining cryptocurrency
- 3 ways to obtain

1. Mining : use powerful computers, issues : graphics card demand and generation of e waste
2. Selling goods to owner of CC : public and private key, untraceable, money laundering and terror finance, government will be deprived of taxes; **Sovereign function of money creation**

Blockchain : Cryptocurrency

1. Exchanging legal tender : no protection under FEMA act, tax haven issues, illegal trade and terror finance

- Which organization helps curb terror finance?

Bitcoin — asymmetric cryptography

private key ⟷ public key (math link)

private key → secret

public key → public

· any change to ledger — pvt key used → creates a digital signature → verified by public key

One way function

· Mathematical principle

· to derive pvt key from Public key, astronomical time needed.

Pvt key → Public key (easy)

Public key → Pvt key (near impossible)

Clarity for UPSC by Dr. Shivin

Ed Sarthi Your Own Mentor

Shor's Quantum Algorithm

- Break the one way function.
- ∴ Large enough Quantum Computer → digital signature can be falsified.

Bitcoin

- decentralised generation of pvt and public key
- for transaction
  - use pvt key — validate transaction → spend your BTC.

Quantum computer — public key → pvt key

A ⟶ B

transaction reveals public key
↓
till it is not mined
↓
vulnerable to Quantum attack.

Mining → competition b/w validators.
→ Quantum co. takes over.

↓

during this time, your a/c is vulnerable

↓ current mining time = 10 min

↓ Quantum Shor's algo predicted time ↓ 30min-8h.

Web 3.0

- Web 1.0, also called the Static Web, which enabled easy access to information. However, the information was largely disorganised and difficult to navigate.
- The Social Web, or Web 2.0, made the internet a lot more interactive thanks to advancements in web technologies like Javascript, HTML5, CSS3, etc., which enabled startups to build interactive web platforms such as YouTube, Facebook, Wikipedia and many more.

Web 3.0

- Facebook.com : different for all of us
- Brainstorm : Web2.0 problems?
- Web 3.0 : third stage in internet evolution

Web 3.0

- Web 2.0 : you are the product
- Web 3.0 : you are the owner
- new iteration of the World Wide Web which would incorporate concepts such as decentralization, distributed ledger such as block chain technologies, and token-based economics.
- How?

THREE STAGES OF INTERNET CONSUMPTION

	Web1	Web2	Web3
Time period*	1990-2005	2005-till date	2021
Where data is stored	Server's file system	On-premise/Cloud	Blockchain, distributed across multiple networks
Examples	Static web pages	User generated content like Social media, and web applications like e-commerce etc...	NFTs, cryptocurrency transaction
Who owns data	Companies running the webpages	Companies that host application, cloud service providers	No one owns the data
Transacting	No transaction possible	Payment gateways for currency transactions	Transaction happens using crypto tokens

Web 3.0 explained

- Its like a big torrent network
- Data is stored in blockchain over many computers around the world
- Token based economics : Example : odyssey website : alternative to YouTube : creators earn library tokens for making others watch
- Post on many computers around the world : cannot be taken down or censored.

Web 3.0 explained

- Who runs it : DAO for every company
- Decentralised autonomous organization : no CEO, no one controlling authority
- Most tokens : decision on rules etc
- Anonymized identity

Web 3.0 advantages

- Decentralization: core tenet of Web 3.0; could be stored in multiple locations simultaneously; break down massive databases currently held by internet giants like Meta and Google;
- Democratize net: Greater control to user; data security;
- Trustless and permissionless: run on decentralized peer-to-peer networks – Trustless (i.e., the network will allow participants to interact directly without going through a trusted intermediary); permissionless (meaning that anyone can participate without authorization from a governing body);

Web 3.0 advantages

- dApp- Decentralized applications (dApps): applications built on blockchain; use smart contracts to enable service delivery in a programmatic approach;
- DeFi - Decentralized finance (DeFi): use blockchain for enabling financial services
- DAOs - providing structure and governance in a decentralized approach.

Web 3.0 advantages

- Artificial intelligence (AI) and machine learning: Computers will be able to understand information similarly to humans; will use machine learning, - data and algorithms to imitate how humans learn;
- Others: NFT, Crypto currency;

Web 3.0 challenges

- Consensus approach: slow development, decisions;
- Regulation
- Data privacy
- Malafide use
- Not decentralised: controlled by capitalists, big firms;
- Business models: averse to decentralization; stakeholders, assets drive economy;
- Finances: acceptance and illegality of crypto e.g. torrent

Non Fungible Tokens

- Digital identifier, similar to a certificate of ownership that represents a digital or physical asset; financial security consisting of digital data stored in a blockchain, a form of distributed ledger;
- NFT market grew exponentially from 2020–2021: the trading of NFTs in 2021 increased to more than $17 billion, up by 21,000% over 2020's total of $82 million; Work entitled Merge by artist Pak was one the most expensive NFT, with an auction price of US$91.8 million.
- What is fungibility?

Fungibility

- Fungibility means subdivision into units as well as mutual substitution
- Example : Rs. 2000 note.
- Subdivided : yes, 500 x 4 notes
- Mutual substitution : yes, both have the same value
- Other cases :
- 1 kg gold bar
- 100 gm diamond
- NFT : not possible

NFT

- NFT is a Digital file such as photo file(JPEG),Animated image(GIF),music file (MP3) etc. Stored using blockchain Technology.
- They cannot be sub divided and/or their individual sub-units can not be exchanged with one another. Because their values are different based on buyer’s preference. So NFTs are non- fungible.
- How does it work?

NFT : working explained (in class)

NFT

- Advantages :
- Multimedia creators : trading of digital assets
- Easier to verify opwnership records
- Paper less administrations : digital torage of tickets / records
- Challenges :
- mass hysteria of investment bubble might collapse leading to bankruptcy of gullible citizens
- Money laundering and terror finance : same as bitcoin
- Electricity consumption and environmental challenges

Internet Of Things

- It is a seamless connected network of embedded objects/ devices, with identifiers, in which Machine to Machine (M2M) communication without any human intervention is possible using standard and interoperable communication protocols.
- In general terms, it includes any object or thing that can be connected to an Internet network, from factory equipment and cars to mobile devices and smart watches.
- Phones, Tablets and PCs are not included as part of IoT.

Internet Of Things

- It specifically comes to mean **connected things that are equipped with sensors, software, and other technologies** that allow them to transmit and receive data.
- It is widely being used to create **smart infrastructure in various verticals such as Power, Automotive, Safety and Surveillance, Remote Health Management, Agriculture, Smart Homes and Smart Cities** etc, using connected devices.

How does it work?

- An IoT ecosystem consists of web- enabled smart devices use embedded systems, such as processors, sensors and communication hardware, to collect, send and act on data they acquire from their environments.
- These share the sensor data they collect by connecting to an IoT gateway or other edge device where data is sent to the cloud to be analyzed locally.

How does it work?

- Sometimes, these devices communicate with other related devices and act on the information they get from one another.
- The devices do most of the work without human intervention, although people can interact with the devices for instance, to set them up, give them instructions or access the data.

Internet of Things Uses

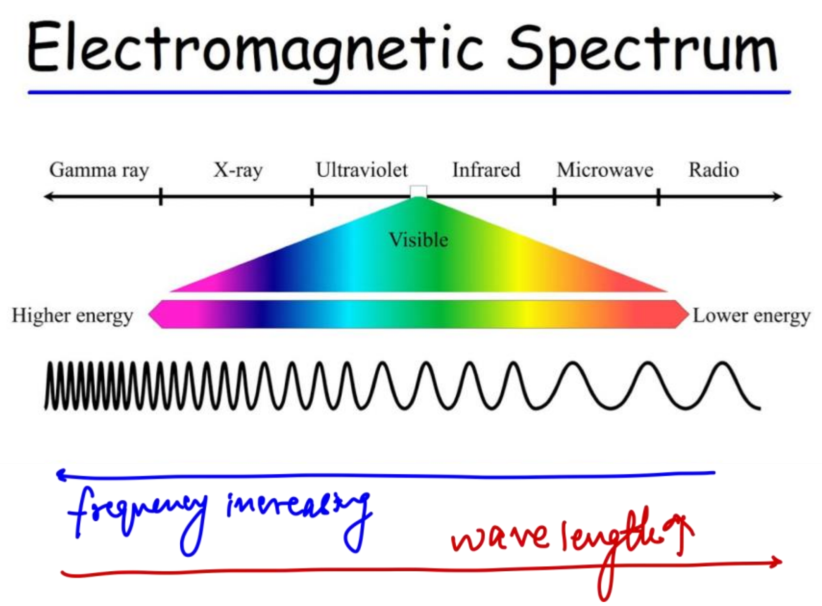
Significance and challenges

- IoT encourages the communication between devices, also known as M2M communication.
- M2M communication helps to maintain transparency in the processes. It leads to uniformity in the tasks and maintain the quality of service.
- M2M interaction provides better efficiency, hence, accurate results can be obtained fast.
- Applications of IoT in increased convenience, better management, thereby improving the quality of life.

Significance and challenges

- Challenges of IoT: Data Breach, increased dependence on Technology, Complexity in operation, issue of compatibility in tagging and monitoring, lesser Employment of Menial Staff etc.
- Further challenges : internet consumption and bandwidth issues
- Solutions : Lifi , 5g

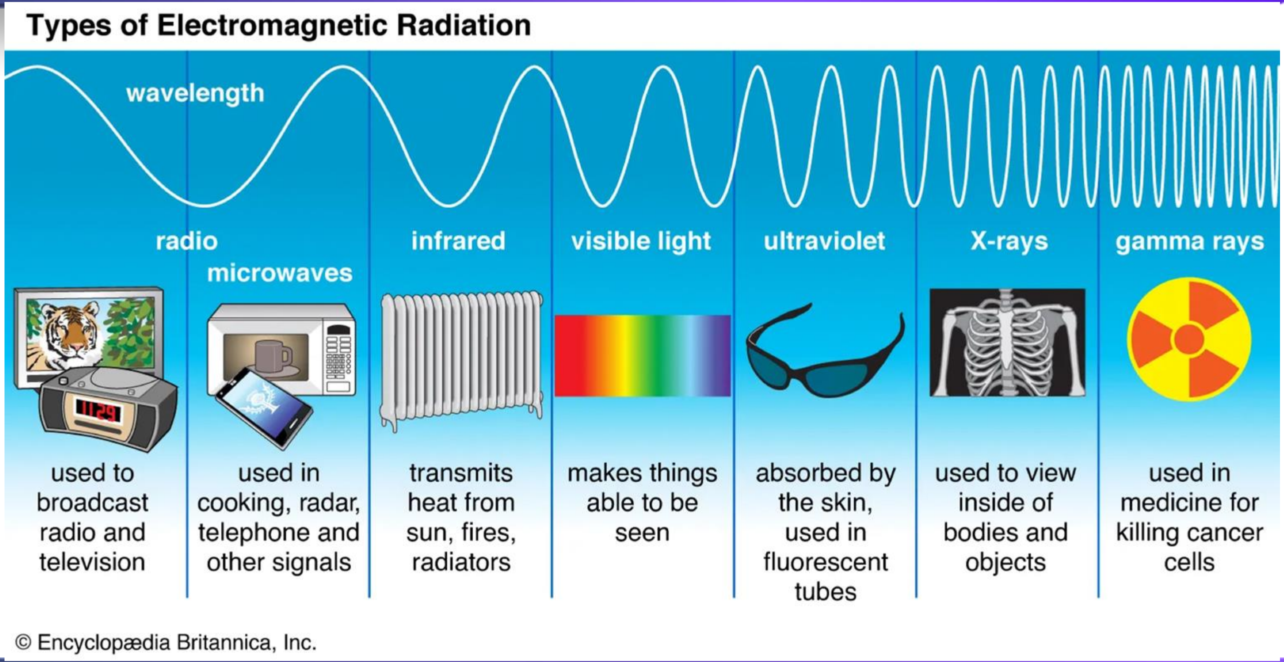
Electromagnetic Spectrum Explained



Full 'spectrum' of radiation emitted from a star.

$$E = h\frac{c}{\lambda} = h\nu$$

Electromagnetic Spectrum Explained



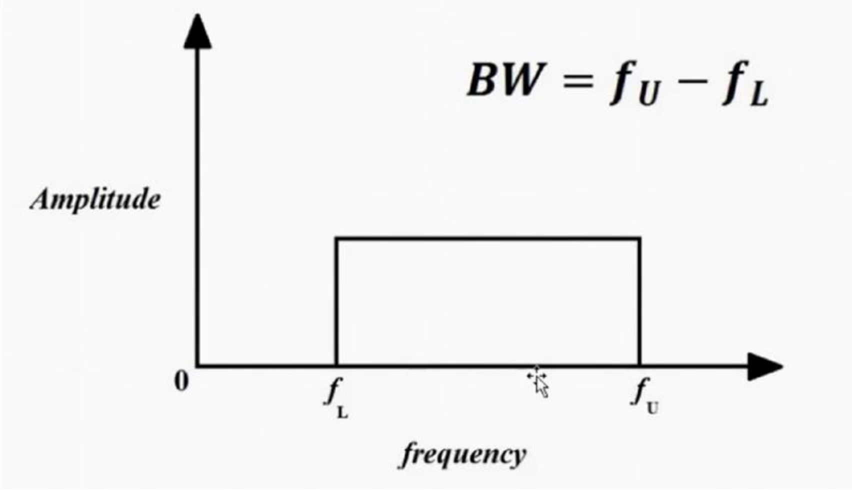
Electromagnetic Spectrum Explained

EM Wave	Frequency Range
Extremely high freqenecies (EHF)	30 – 300 GHz
Super high frequencies (SHF)	3 – 30 GHz
Ultra high frequencies (UHF)	300 MHz – 3 GHz
Very high frequencies (VHF)	30 – 300 MHz
High frequencies (HF)	3 – 30 MHz
Medium frequencies (MF)	300 KHz – 3 MHz
Low Frequencies (LF)	30 – 300 KHz
Very low frequencies (VLF)	3 – 30 KHz
Voice frequencies (VF)	300 – 3000 Hz
Extremely low frequencies (ELF)	30 – 300 Hz

Electromagnetic Spectrum Explained

What is Bandwidth ?

- *Bandwidth* is the *portion of the electromagnetic spectrum occupied by a signal.*
- It is the *frequency range over which signal is transmitted.*



LIFI technology

- LiFi is a wireless optical networking technology, which uses light emitting diodes (LEDs) to transmit data.
- It makes a LED light bulb emit pulses of light that are undetectable to human eye and within those emitted pulses, data can travel to and from receivers.

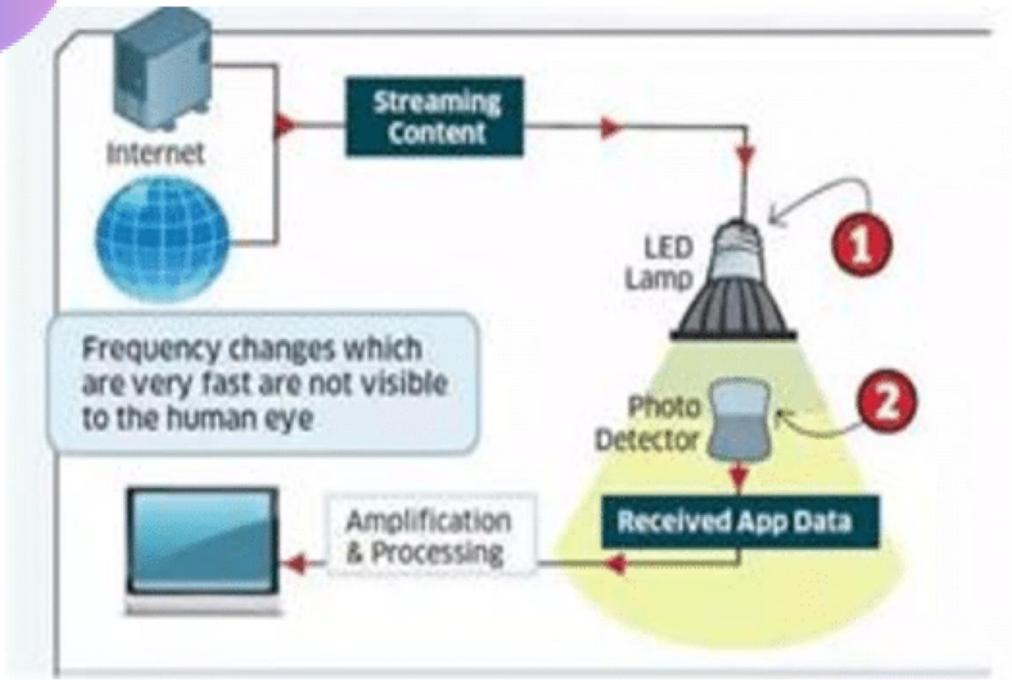
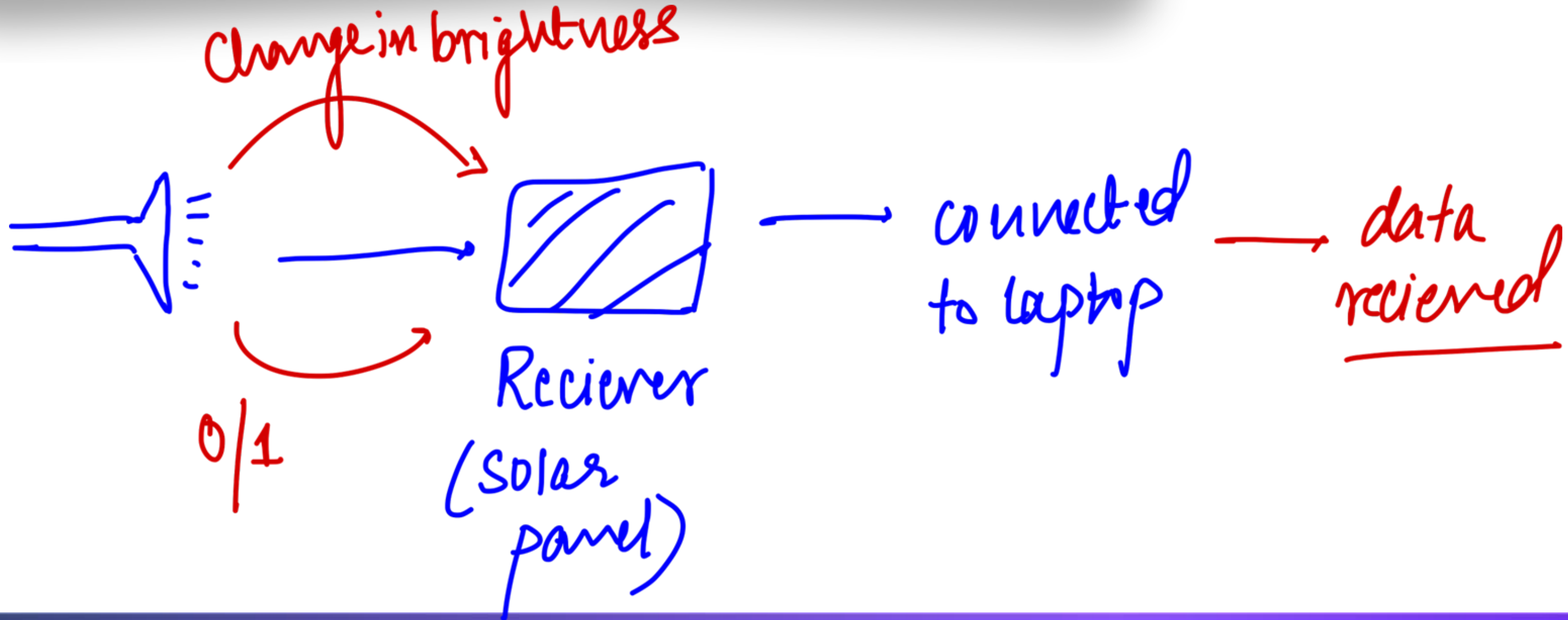


Figure 2: Procedure of Li-fi Technology

LIFI explained



LIFI explained : visible light

Metaverse

- Metaverse projects refer to blockchain-based games and applications that are set in virtual worlds.
- Such games follow a play-to-earn model, where gamers are rewarded for being invested in the game.
- Metaverse can be understood as a combination of multiple elements of technology, including virtual reality, augmented reality and video where users "live" within a digitally enhanced surrounding.

Metaverse

- **These technologies** are currently being developed by Apple, Google, Amazon, and Microsoft in addition to Facebook.
- Metaverse uses the following technologies : (**extended realities)**
- Virtual Reality
- Augmented Reality
- Hologram
- Avatar
- Let's see a video on how it works

Metaverse : Technologies

- **Virtual Reality (VR):** VR can be understood as the use of computer modelling and simulation that enables a person to interact with an artificial three-dimensional (3-D) visual or other sensory environment. E.g., games like World of Warcraft. **VR closes the world,** and transposes an individual, providing complete immersion experience.

Metaverse : Technologies

- **Augmented Reality (AR):** It is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. E.g., games like Pokémon Go.
- **Hologram:** Holograms are virtual three-dimensional images created by the interference of light beams that reflect real physical objects.

Metaverse : Technologies

- **Avatar:** An avatar in the metaverse is a representation of an individual in the virtual world, this digital avatar enables the person to function like an actual human being in a digitally created world.
- **Platform/Developer/User distinction:** In metaverse, digital engagement will get very personal and tailormade for the user. In this context, it is important that role played by the platform, the developers and the users is clearly understood.

Metaverse : Technologies : Images

5G technology

- 5G (the fifth generation of cellular networks) is designed to improve network connections by **addressing the legacy issues of speed, latency and utility,** which the earlier/ current generation of mobile networks could not address.
- **5G operates at higher frequencies** to offer a new kind of network that is **designed to connect virtually everyone and everything together** including machines, objects, and devices.

5G technology

- It will also have an **enhanced throughput to handle more simultaneous connections** at a time than current-generation networks.
- 5G **mainly works in 3 bands,** namely low, mid and high frequency spectrum.

3G vs 4G vs 5G vs 6G

	3G	4G	5G	6G
Deployment	2004-06	2006-10	2020	2028-2030
Bandwidth	2 mbps	200 mbps	>1 gbps	1 tbps
Latency	100-150 millisecond	20-30 millisecond	<10 millisecond	<1 microsecond
Average Speed	144 kbps	25 mbps	200-400 mbps	About 50 times faster than 5G

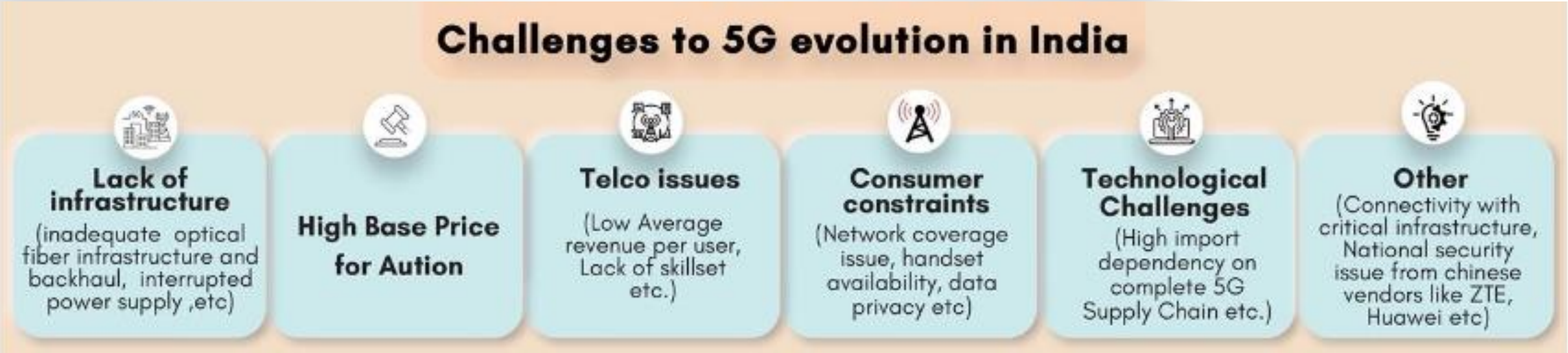
Clarity for UPSC by Dr. Shivin

Ed Sarthi Your Own Mentor

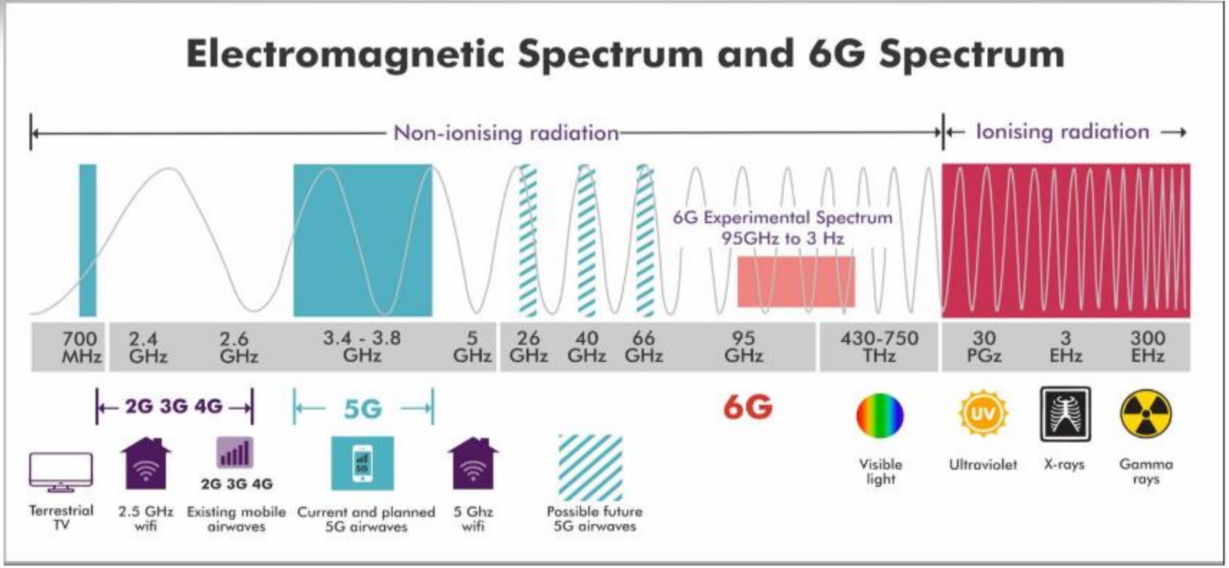
5G technology

Spectrum	Uses/Advantage	Limitations
Low Band	• Great promise in terms of coverage and speed of internet and data exchange, the maximum speed is limited to 100 Mbps (Megabits per second) • For commercial cellphone users who may not have specific demands for every high speed internet.	• May not be optimal for specialised needs of the industry.
Mid-Band	• High speeds compared to the low band. • May be used by industries and specialised factory building captive networks.	In terms of coverage area and penetration of signals.
High-band	• Offers the highest speed (as high as 20 gbps) (giga bits per seconds) of all the three bans.	• Extremely limited coverage and signal penetration strength.

5G technology : challenges in India



5G and EM spectrum

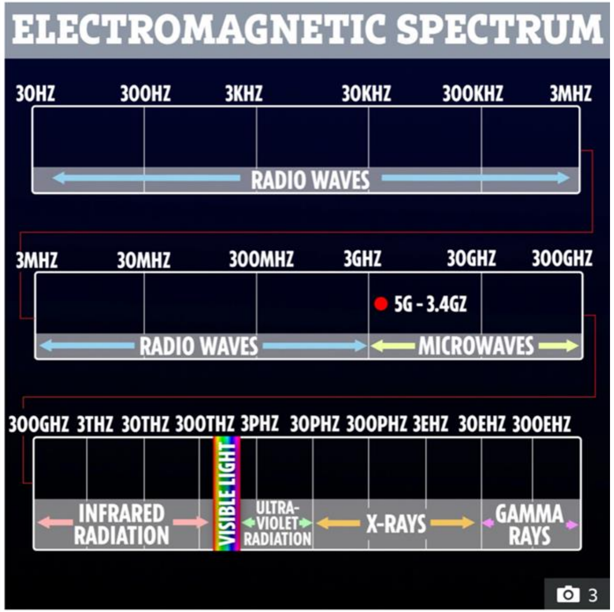


5G and EM spectrum

⌃ Electromagnetic spectrum

Microwaves occupy a place in the electromagnetic spectrum with frequency above ordinary radio waves, and below infrared light:

Electromagnetic spectrum			
Name	Wavelength	Frequency (Hz)	Photon energy (eV)
Gamma ray	< 0.01 nm	> 30 EHz	> 124 keV
X-ray	0.01 nm – 10 nm	30 EHz – 30 PHz	124 keV – 124 eV
Ultraviolet	10 nm – 400 nm	30 PHz – 750 THz	124 eV – 3 eV
Visible light	400 nm – 750 nm	750 THz – 400 THz	3 eV – 1.7 eV
Infrared	750 nm – 1 mm	400 THz – 300 GHz	1.7 eV – 1.24 meV
Microwave	1 mm – 1 m	300 GHz – 300 MHz	1.24 meV – 1.24 µeV
Radio	≥ 1 m	≤ 300 MHz	≤ 1.24 µeV



5G is a very low-frequency form of radiation – far below visible light and infrared

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- VoLTE is a **technology update to the LTE protocol** used by mobile phone networks.
- **Under LTE,** the infrastructure of telecom players **only allows transmission of data while voice calls are routed to their older 2G or 3G networks.**
- This is why, under LTE, you cannot access your 4G data services while on a call. This leads to problems

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- Such as slow internet speeds and poor voice clarity.
- LTE is commonly marketed as 4G LTE.
- **VoLTE allows voice calls to be 'packaged' and carried through LTE networks**. This would mean 4G data accessibility even during calls.

Dark Net

- Imagine you get an email containing your bank statement :
- It is on the internet
- It is not accessible to me.
- Also known as Dark Web, it is that part of Internet which cannot be accessed through traditional search engines like Google nor is it accessible by normal browsers like Chrome or Safari.

Dark Net

- It generally uses non-standard communication protocols which make it inaccessible to internet service providers (ISPs) or government authorities.
- Content on Dark Net is encrypted and requires specific browser such as TOR (The Onion Router) browser to access those pages.

Dark Net

- Dark Net itself is only a part of Deep Web that is a broader concept, which includes sites that are protected by passwords.
- Part of internet that is readily available to general public and searchable on standard search engines is called as Surface Web.
- It is used by journalists and citizens working in oppressive regimes, researchers and students to do research on sensitive topics, law enforcement agencies etc.

Dark Net

- Concerns over its use: Anonymity, Haven for illicit activity, Privacy and ethical concerns, drug dealing, communication by terrorists etc.